Hook
Siller, DS 4002

The Case of the Unhappy Passengers:
Investigating Airline Reviews Using Sentiment Analysis

Fig. 1: Airline Passenger Feedback Image. Image from: *Dreamstime*, https://www.dreamstime.com/photos-images/airline-feedback.html (accessed 5/3/2026).

Background: It's early morning at the airport. You're waiting at the gate, watching passengers slowly gather before boarding. Some look relaxed, sipping coffee and scrolling through their phones. Others look frustrated already, even before stepping onto the plane. A few days later, those same passengers are leaving reviews online. One describes their flight as amazing, praising the crew and comfort. Another complains about cramped seating and poor service. Others fall somewhere in between, calling the experience average or forgettable. As you scroll through review after review, a pattern starts to stand out. Passengers who mention Business Class seem more positive, while Economy passengers appear more mixed. Still, it is difficult to tell if this pattern is real or just coincidence. With thousands of reviews across multiple airlines, individual impressions are not enough to draw a reliable conclusion. Airlines rely heavily on customer satisfaction, and these reviews influence future travelers as well as business decisions. If passengers in higher classes are consistently more satisfied, that could justify differences in pricing, service, and resource allocation. Instead of relying on assumptions, you decide to approach the question using data.

Task and Deliverable: You are a data scientist working with a team analyzing airline customer experience. Your task is to determine whether passenger class influences the sentiment of airline reviews. Using a dataset of over 8,000 reviews from major international airlines in 2023, you will classify each review as positive, negative, or neutral using a pre-trained sentiment analysis model. You will then explore patterns in the data through visualizations and compare sentiment across Economy and Business Class passengers. Finally, you will perform a chi-squared test to determine whether any observed differences are statistically significant. Your goal is to answer the question: is there a statistically significant difference in customer sentiment between Economy and Business Class passengers? Your analysis will be considered successful if you are able to generate sentiment labels, produce clear visualizations, perform and interpret the statistical test, and draw a well-supported conclusion based on your results. Everything needed to complete this case study is here: https://github.com/wtym27/DS4002-CS3/blob/main/README.md.